

Experiment-5

Aim: Perform Social media analysis using Cassandra.

1.start ubuntu,open terminal

2.install docker

sudo apt update

sudo apt install docker.io -y

3.Start Docker:

sudo systemctl start docker

sudo systemctl enable docker

(Optionally add your user to docker group)

sudo usermod -aG docker student

Then restart your system

4. — Pull Cassandra image

docker pull cassandra

5.— Run Cassandra container

docker run --name cassandra1 -d -p 9042:9042 cassandra

Check if it is running:

docker ps

6.— Enter Cassandra container

docker exec -it cassandra1 bash

Now open CQL shell: cqlsh

if it is showing error the use this cqlsh 127.0.0.1 9042

source code

```
create keyspace social_media with replication = {'class':'SimpleStrategy','replication_factor':1};
```

```
use social_media;
```

```
create table social_media.users(  
  user_id int primary key,  
  username text,  
  email text,  
  registration_date timestamp  
);
```

```
insert into social_media.users(user_id,username,email,registration_date)  
values(1,'user1','user1@example.com',toTimestamp(now()));
```

```
insert into social_media.users(user_id,username,email,registration_date)
values(2,'user2','user2@example.com',toTimestamp(now()));
insert into social_media.users(user_id,username,email,registration_date)
values(3,'user3','user3@example.com',toTimestamp(now()));
insert into social_media.users(user_id,username,email,registration_date)
values(4,'user4','user4@example.com',toTimestamp(now()));
insert into social_media.users(user_id,username,email,registration_date)
values(5,'user5','user5@example.com',toTimestamp(now()));

select * from users;
```

```
CREATE TABLE social_media.user_post_activity (
  user_id int,
  post_id int,
  activity_type text,
  activity_timestamp timestamp,
  PRIMARY KEY (user_id, post_id)
);
```

```
INSERT INTO social_media.user_post_activity (user_id, post_id,
activity_type,activity_timestamp)VALUES (2, 101, 'like', toTimestamp(now()));
```

```
INSERT INTO social_media.user_post_activity (user_id, post_id,
activity_type,activity_timestamp)VALUES (3, 102, 'comment', toTimestamp(now()));
```

```
INSERT INTO social_media.user_post_activity (user_id, post_id,
activity_type,activity_timestamp)VALUES (4, 103, 'share', toTimestamp(now()));
```

```
INSERT INTO social_media.user_post_activity (user_id, post_id, activity_type,
activity_timestamp)VALUES (5, 101, 'like', toTimestamp(now()));
```

```
INSERT INTO social_media.user_post_activity (user_id, post_id, activity_type,
activity_timestamp)VALUES (1, 104, 'comment', toTimestamp(now()));
```

```
CREATE TABLE social_media.posts (
  post_id int PRIMARY KEY,
  user_id int,
  content text,
  created_at timestamp
);
```

```
INSERT INTO social_media.posts (post_id, user_id, content, created_at)VALUES (101, 1, 'Just
joined this awesome social platform!', toTimestamp(now()));
```

```
INSERT INTO social_media.posts (post_id, user_id, content, created_at)VALUES (102, 2, 'Does anyone have book recommendations for AI?', toTimestamp(now()));
```

```
INSERT INTO social_media.posts (post_id, user_id, content, created_at)VALUES (103, 3, 'Throwback to my first coding project in Python!', toTimestamp(now()));
```

```
INSERT INTO social_media.posts (post_id, user_id, content, created_at)VALUES (104, 4, 'Morning thoughts: Coffee > Code or Code > Coffee?', toTimestamp(now()));
```

```
INSERT INTO social_media.posts (post_id, user_id, content, created_at)VALUES (105, 5, 'Who else is excited for the weekend hackathon?!', toTimestamp(now()));
```

```
SELECT * FROM social_media.users;
```

```
SELECT * FROM social_media.posts WHERE user_id = 1 ALLOW FILTERING;
```

```
SELECT * FROM social_media.user_post_activity WHERE user_id = 3;
```

```
SELECT * FROM social_media.user_post_activity WHERE post_id = 101 AND activity_type = 'like' ALLOW FILTERING;
```

```
SELECT post_id, activity_type FROM social_media.user_post_activity WHERE user_id = 2;
```

...

week-6

Aim:buyer event analytics using cassandra on suitable product sales data

source code:

```
create keyspace buyer_events with replication = {'class' : 'SimpleStrategy','replication_factor':'1'};
```

```
create table buyers(  
    buyer_id int primary key,  
    username text,  
    email text,  
    address text);
```

```
insert into buyers(buyer_id,username,email,address)  
    values(1,'user1','user1@example.com','hyd');  
cqlsh:buyer_events> insert into buyers(buyer_id,username,email,address)  
    values(2,'user2','user2@example.com','mum');  
cqlsh:buyer_events> insert into buyers(buyer_id,username,email,address)  
    values(3,'user3','user3@example.com','pun');  
cqlsh:buyer_events> insert into buyers(buyer_id,username,email,address)  
    values(4,'user4','user4@example.com','hyd');  
cqlsh:buyer_events> insert into buyers(buyer_id,username,email,address)  
    values(5,'user5','user5@example.com','pun');  
insert into buyers(buyer_id,username,email,address)  
    values(6,'user6','user6@example.com','mum');  
select * from buyers;
```

```
create table products(  
    product_id int primary key,  
    name text,  
    category text,  
    price int);
```

```
cqlsh:buyer_events> insert into products(product_id,name,category,price)  
    values (1, 'coffee maker', 'appliances', 60);  
cqlsh:buyer_events> insert into products (product_id, name, category, price)  
    values (2, 'phone', 'electronics', 300);  
cqlsh:buyer_events> insert into products (product_id, name, category, price)  
    values (3, 'led tv', 'electronics', 600);  
cqlsh:buyer_events> insert into products (product_id, name, category, price)  
    values (4,'running shoes', 'apparel', 100);  
cqlsh:buyer_events> insert into products (product_id, name, category, price)  
    values (5,'oven', 'appliances', 500);  
cqlsh:buyer_events> select * from products;
```

```
create table purchase_history(  
    transactionID int primary key,  
    buyer_id int,  
    product_id int,  
    quantity int,  
    total_amount int,  
    purchase_date timestamp);
```

```
insert into  
purchase_history(transactionID,buyer_id,product_id,quantity,total_amount,purchase_date)
```

```
cqlsh:buyer_events> insert into  
purchase_history(transactionID,buyer_id,product_id,quantity,total_amount,purchase_date)  
... values(2,1,4,4,400,toTimestamp(now()));
```

```
cqlsh:buyer_events> insert into  
purchase_history(transactionID,buyer_id,product_id,quantity,total_amount,purchase_date)  
... values(3,2,4,1,100,toTimestamp(now()));
```

```
cqlsh:buyer_events> insert into  
purchase_history(transactionID,buyer_id,product_id,quantity,total_amount,purchase_date)  
... values(4,2,1,3,180,toTimestamp(now()));
```

```
cqlsh:buyer_events> insert into  
purchase_history(transactionID,buyer_id,product_id,quantity,total_amount,purchase_date)  
... values(5,3,3,4,2400,toTimestamp(now()));
```

```
cqlsh:buyer_events> insert into  
purchase_history(transactionID,buyer_id,product_id,quantity,total_amount,purchase_date)  
... values(6,4,2,2,600,toTimestamp(now()));
```

QUERIES :

1. RETRIEVE BUYER'S PURCHASE HISTORY

```
select * from purchase_history where buyer_id = 1 allow filtering;
```

2. FIND TOTAL NUMBER OF PRODUCTS

```
select count(product_id) from products;
```

3. RETRIEVE TOTAL SPENDING BY A BUYER

select sum(total_amount) from purchase_history where buyer_id = 2 allow filtering;

4. RETRIEVE PRODUCTS BY CATEGORY

select * from products where category = 'electronics' allow filtering;

5. FIND THE MAXIMUM PRICE

Select max(price) from products;

6. TO FIND TOTAL NUMBER OF BUYERS

select count(buyer_id) from buyers;

7. FIND AVERAGE PRICE OF PRODUCTS OF A PARTICULAR CATEGORY

select avg(price) from products where category='electronics' allow filtering;

8. SELECT PRODUCTS IN A PARTICULAR PRICE RANGE

select * from products where price>=100 and price<=400 allow filtering;

9. TO FIND NUMBER OF BUYERS FROM A SPECIFIC LOCATION

select count(buyer_id) from buyers where address='pun' allow filtering;

10. TO FIND ALL THE PURCHASES OF A PARTICULAR PRODUCT

select * from purchase_history where product_id = 4 allow filtering;

11. TO FIND TOTAL AMOUNT OBTAINED THROUGH SALES OF A PARTICULAR PRODUCT

```
select sum(total_amount) from purchase_history where product_id = 4 allow filtering;
```